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KOMATSU D155A-5 SHOP MANUAL

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SHOP MANUAL

KOMATSU

D155A-5

Machine model	Serial number
D155A-5	**65001 and up**

- This shop manual may contain attachments and optional equipment that are not available in your area. Please consult your local Komatsu distributor for those items you may require. Materials and specifications are subject to change without notice.
- D155A-5 mounts the SA6D140E-2 engine.
  For details of the engine, see the 6D140-2 Series Engine Shop Manual.

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SAFETY
SAFETY NOTICE

IMPORTANT SAFETY NOTICE

Proper service and repair is extremely important for safe machine operation. The service and repair techniques recommended by Komatsu and described in this manual are both effective and safe. Some of these techniques require the use of tools specially designed by Komatsu for the specific purpose.

To prevent injury to workers, the symbol ⚠ is used to mark safety precautions in this manual. The cautions accompanying these symbols should always be followed carefully. If any dangerous situation arises or may possibly arise, first consider safety, and take the necessary actions to deal with the situation.

GENERAL PRECAUTIONS

Mistakes in operation are extremely dangerous. Read the Operation and Maintenance Manual carefully BEFORE operating the machine.

1. Before carrying out any greasing or repairs, read all the precautions given on the decals which are fixed to the machine.
2. When carrying out any operation, always wear safety shoes and helmet. Do not wear loose work clothes, or clothes with buttons missing.
   - Always wear safety glasses when hitting parts with a hammer.
   - Always wear safety glasses when grinding parts with a grinder, etc.
3. If welding repairs are needed, always have a trained, experienced welder carry out the work. When carrying out welding work, always wear welding gloves, apron, hand shield, cap and other clothes suited for welding work.
4. When carrying out any operation with two or more workers, always agree on the operating procedure before starting. Always inform your fellow workers before starting any step of the operation. Before starting work, hang UNDER REPAIR signs on the controls in the operator's compartment.
5. Keep all tools in good condition and learn the correct way to use them.
6. Decide a place in the repair workshop to keep tools and removed parts. Always keep the tools and parts in their correct places. Always keep the work area clean and make sure that there is no dirt or oil on the floor. Smoke only in the areas provided for smoking. Never smoke while working.

PREPARATIONS FOR WORK

7. Before adding oil or making any repairs, park the machine on hard, level ground, and block the wheels or tracks to prevent the machine from moving.
8. Before starting work, lower blade, ripper, bucket or any other work equipment to the ground. If this is not possible, insert the safety pin or use blocks to prevent the work equipment from falling. In addition, be sure to lock all the control levers and hang warning signs on them.
9. When disassembling or assembling, support the machine with blocks, jacks or stands before starting work.
10. Remove all mud and oil from the steps or other places used to get on and off the machine. Always use the handrails, ladders or steps when getting on or off the machine. Never jump on or off the machine. If it is impossible to use the handrails, ladders or steps, use a stand to provide safe footing.

PRECAUTIONS DURING WORK

11. When removing the oil filler cap, drain plug or hydraulic pressure measuring plugs, loosen them slowly to prevent the oil from spurting out.
    Before disconnecting or removing components of the oil, water or air circuits, first remove the pressure completely from the circuit.
12. The water and oil in the circuits are hot when the engine is stopped, so be careful not to get burned.
    Wait for the oil and water to cool before carrying out any work on the oil or water circuits.
13. Before starting work, remove the leads from the battery. Always remove the lead from the negative (–) terminal first.
14. When raising heavy components, use a hoist or crane.
    Check that the wire rope, chains and hooks are free from damage.
    Always use lifting equipment which has ample capacity.
    Install the lifting equipment at the correct places.
    Use a hoist or crane and operate slowly to prevent the component from hitting any other part.
    Do not work with any part still raised by the hoist or crane.
15. When removing covers which are under internal pressure or under pressure from a spring, always leave two bolts in position on opposite sides. Slowly release the pressure, then slowly loosen the bolts to remove.
16. When removing components, be careful not to break or damage the wiring. Damaged wiring may cause electrical fires.
17. When removing piping, stop the fuel or oil from spilling out. If any fuel or oil drips onto the floor, wipe it up immediately. Fuel or oil on the floor can cause you to slip, or can even start fires.
18. As a general rule, do not use gasoline to wash parts. In particular, use only the minimum of gasoline when washing electrical parts.
19. Be sure to assemble all parts again in their original places.
    Replace any damaged parts with new parts.
    - When installing hoses and wires, be sure that they will not be damaged by contact with other parts when the machine is being operated.
20. When installing high pressure hoses, make sure that they are not twisted. Damaged tubes are dangerous, so be extremely careful when installing tubes for high pressure circuits. Also, check that connecting parts are correctly installed.
21. When assembling or installing parts, always use the specified tightening torques. When installing protective parts such as guards, or parts which vibrate violently or rotate at high speed, be particularly careful to check that they are installed correctly.
22. When aligning two holes, never insert your fingers or hand. Be careful not to get your fingers caught in a hole.
23. When measuring hydraulic pressure, check that the measuring tool is correctly assembled before taking any measurements.
24. Take care when removing or installing the tracks of track-type machines.
    When removing the track, the track separates suddenly, so never let anyone stand at either end of the track.

FOREWORD
GENERAL

This shop manual has been prepared as an aid to improve the quality of repairs by giving the serviceman an accurate understanding of the product and by showing him the correct way to perform repairs and make judgements. Make sure you understand the contents of this manual and use it to full effect at every opportunity.

This shop manual mainly contains the necessary technical information for operations performed in a service workshop. For ease of understanding, the manual is divided into the following chapters; these chapters are further divided into the each main group of components.

STRUCTURE AND FUNCTION

This section explains the structure and function of each component. It serves not only to give an understanding of the structure, but also serves as reference material for troubleshooting.
In addition, this section may contain hydraulic circuit diagrams, electric circuit diagrams, and maintenance standards.

TESTING AND ADJUSTING

This section explains checks to be made before and after performing repairs, as well as adjustments to be made at completion of the checks and repairs.
Troubleshooting charts correlating "Problems" with "Causes" are also included in this section.

DISASSEMBLY AND ASSEMBLY

This section explains the procedures for removing, installing, disassembling and assembling each component, as well as precautions for them.

MAINTENANCE STANDARD

This section gives the judgment standards for inspection of disassembled parts.
The contents of this section may be described in STRUCTURE AND FUNCTION.

OTHERS

This section mainly gives hydraulic circuit diagrams and electric circuit diagrams.
In addition, this section may give the specifications of attachments and options together.

NOTICE

The specifications contained in this shop manual are subject to change at any time and without any advance notice. Use the specifications given in the book with the latest date.

HOW TO READ THE SHOP MANUAL

VOLUMES

Shop manuals are issued as a guide to carrying out repairs. They are divided as follows:

Chassis volume: Issued for every machine model
Engine volume: Issued for each engine series
Electrical volume: } Each issued as one volume to cover all models
Attachments volume: }

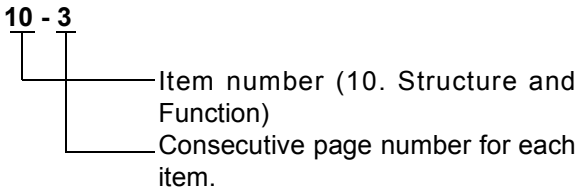
These various volumes are designed to avoid duplicating the same information. Therefore, to deal with all repairs for any model , it is necessary that chassis, engine, electrical and attachment volumes be available.

DISTRIBUTION AND UPDATING

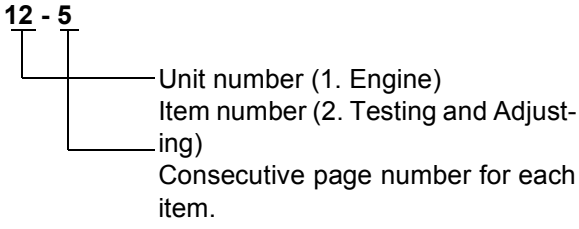
Any additions, amendments or other changes will be sent to KOMATSU distributors. Get the most up-to-date information before you start any work.

FILING METHOD

1. See the page number on the bottom of the page. File the pages in correct order.
2. Following examples show how to read the page number.
   Example 1 (Chassis volume):



   Example 2 (Engine volume):



3. Additional pages: Additional pages are indicated by a hyphen (-) and number after the page number. File as in the example.
   Example:



REVISED EDITION MARK

When a manual is revised, an edition mark ((1)(2)(3)....) is recorded on the bottom of the pages.

REVISIONS

Revised pages are shown in the LIST OF REVISED PAGES next to the CONTENTS page.

SYMBOLS

So that the shop manual can be of ample practical use, important safety and quality portions are marked with the following symbols.

Symbol	Item	Remarks
⚠	Safety	Special safety precautions are necessary when performing the work.
★	Caution	Special technical precautions or other precautions for preserving standards are necessary when performing the work.
	Weight	Weight of parts of systems. Caution necessary when selecting hoisting wire, or when working posture is important, etc.
	Tightening torque	Places that require special attention for the tightening torque during assembly.
	Coat	Places to be coated with adhesives and lubricants, etc.
	Oil, water	Places where oil, water or fuel must be added, and the capacity.
	Drain	Places where oil or water must be drained, and quantity to be drained.

HOISTING INSTRUCTIONS

HOISTING

⚠ Heavy parts (25 kg or more) must be lifted with a hoist, etc. In the **DISASSEMBLY AND ASSEMBLY** section, every part weighing 25 kg or more is indicated clearly with the symbol

- If a part cannot be smoothly removed from the machine by hoisting, the following checks should be made:
  1) Check for removal of all bolts fastening the part to the relative parts.
  2) Check for existence of another part causing interference with the part to be removed.

WIRE ROPES

1) Use adequate ropes depending on the weight of parts to be hoisted, referring to the table below:

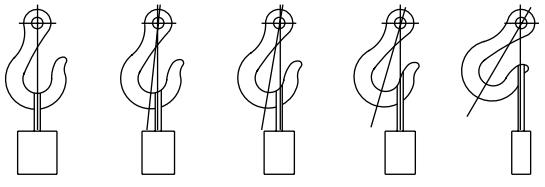
Wire ropes
(Standard "Z" or "S" twist ropes
without galvanizing)

Rope diameter	Allowable load	
mm	kN	tons
10	9.8	1.0
11.5	13.7	1.4
12.5	15.7	1.6
14	21.6	2.2
16	27.5	2.8
18	35.3	3.6
20	43.1	4.4
22.4	54.9	5.6
30	98.1	10.0
40	176.5	18.0
50	274.6	28.0
60	392.2	40.0

★ The allowable load value is estimated to be one-sixth or one-seventh of the breaking strength of the rope used.

2) Sling wire ropes from the middle portion of the hook.

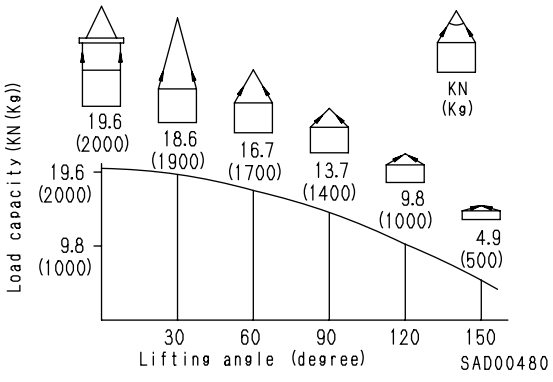
Slinging near the edge of the hook may cause the rope to slip off the hook during hoisting, and a serious accident can result. Hooks have maximum strength at the middle portion.



3) Do not sling a heavy load with one rope alone, but sling with two or more ropes symmetrically wound onto the load.

⚠ Slinging with one rope may cause turning of the load during hoisting, untwisting of the rope, or slipping of the rope from its original winding position on the load, which can result in a dangerous accident.

4) Do not sling a heavy load with ropes forming a wide hanging angle from the hook.
When hoisting a load with two or more ropes, the force subjected to each rope will increase with the hanging angles. The table below shows the variation of allowable load kN {kg} when hoisting is made with two ropes, each of which is allowed to sling up to 9.8 kN {1000 kg} vertically, at various hanging angles.
When two ropes sling a load vertically, up to 19.6 kN {2000 kg} of total weight can be suspended. This weight becomes 9.8 kN {1000 kg} when two ropes make a 120° hanging angle. On the other hand, two ropes are subjected to an excessive force as large as 39.2 kN {4000 kg} if they sling a 19.6 kN {2000 kg} load at a lifting angle of 150°.



METHOD OF DISASSEMBLING, CONNECTING PUSH-PULL TYPE COUPLER

⚠ Before carrying out the following work, release the residual pressure from the hydraulic tank. For details, see TESTING AND ADJUSTING, Releasing residual pressure from hydraulic tank.

⚠ Even if the residual pressure is released from the hydraulic tank, some hydraulic oil flows out when the hose is disconnected. Accordingly, prepare an oil receiving container.

Disconnection

1) Release the residual pressure from the hydraulic tank. For details, see TESTING AND ADJUSTING, Releasing residual pressure from hydraulic tank.

2) Hold adapter (1) and push hose joint (2) into mating adapter (3). (See Fig. 1)
   - ★ The adapter can be pushed in about 3.5 mm.
   - ★ Do not hold rubber cap portion (4).

3) After hose joint (2) is pushed into adapter (3), press rubber cap portion (4) against (3) until it clicks. (See Fig. 2)

4) Hold hose adapter (1) or hose (5) and pull it out. (See Fig. 3)
   - ★ Since some hydraulic oil flows out, prepare an oil receiving container.

Connection

1) Hold hose adapter (1) or hose (5) and insert it in mating adapter (3), aligning them with each other. (See Fig. 4)
   - ★ Do not hold rubber cap portion (4).

2) After inserting the hose in the mating adapter perfectly, pull it back to check its connecting condition. (See Fig. 5)
   - ★ When the hose is pulled back, the rubber cap portion moves toward the hose about 3.5 mm. This does not indicate abnormality, however.

Type 1

Fig. 1

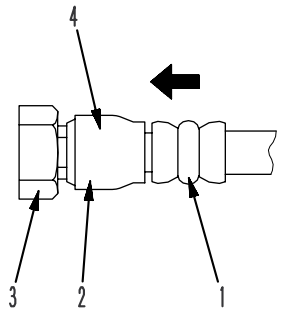


Fig. 2

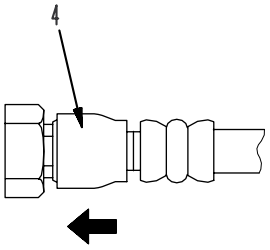


Fig. 3

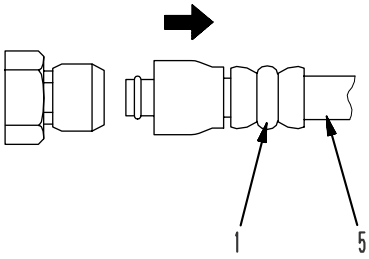


Fig. 4

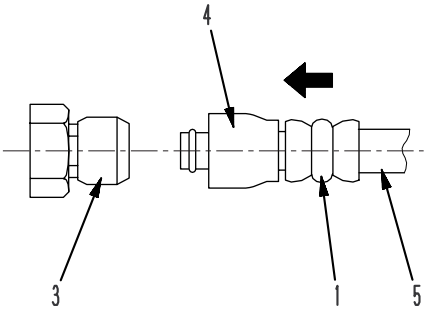
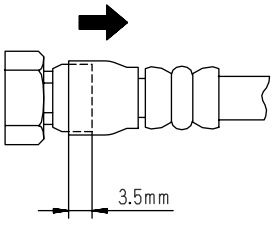
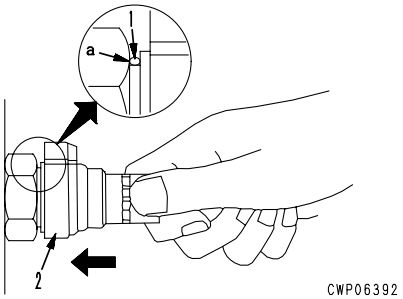
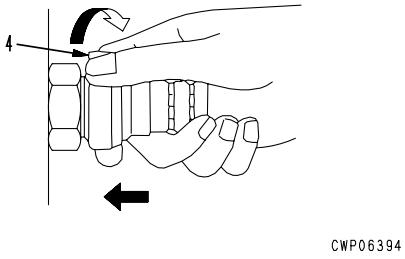


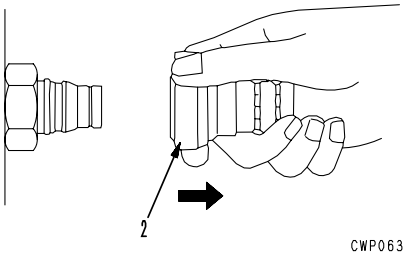
Fig. 5

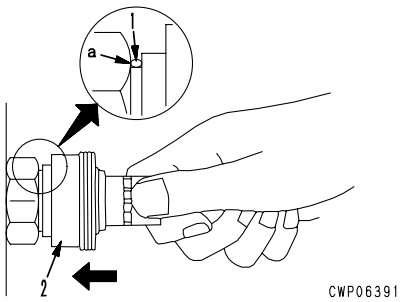


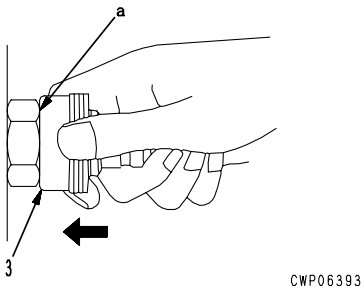
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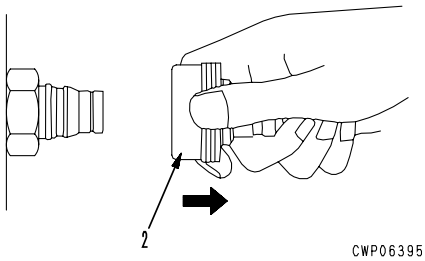
<table>
<tr><th></th><th>Type 2</th><th>Type 3</th></tr>
<tr><td>Disassembly</td><td>
1) Hold the mouthpiece of the tightening portion and push body (2) in straight until sliding prevention ring (1) contacts contact surface a of the hexagonal portion at the male end.


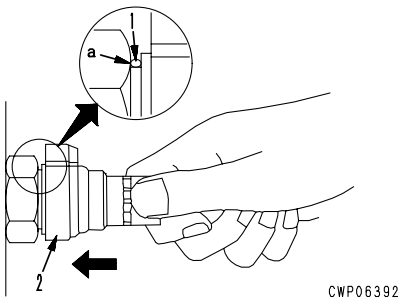
2) Hold in the condition in Step 1), and turn lever (4) to the right (clockwise).


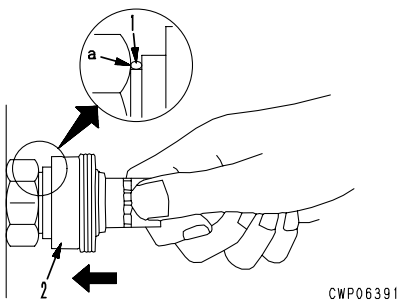
3) Hold in the condition in Steps 1) and 2), and pull out whole body (2) to disconnect it.


</td><td>
1) Hold the mouthpiece of the tightening portion and push body (2) in straight until sliding prevention ring (1) contacts contact surface a of the hexagonal portion at the male end.


2) Hold in the condition in Step 1), and push until cover (3) contacts contact surface a of the hexagonal portion at the male end.


3) Hold in the condition in Steps 1) and 2), and pull out whole body (2) to disconnect it.


</td></tr>
<tr><td>Connection</td><td>
• Hold the mouthpiece of the tightening portion and push body (2) in straight until sliding prevention ring (1) contacts contact surface a of the hexagonal portion at the male end to connect it.


</td><td>
• Hold the mouthpiece of the tightening portion and push body (2) in straight until sliding prevention ring (1) contacts contact surface a of the hexagonal portion at the male end to connect it.


</td></tr>
</table>

COATING MATERIALS

★ The recommended coating materials such as adhesives, gasket sealants and greases used for disassembly and assembly are listed below.
★ For coating materials not listed below, use the equivalent of products shown in this list.

Category	Komatsu code	Part No.	Q'ty	Container	Main applications, features
Adhesives	LT-1A	790-129-9030	150 g	Tube	• Used to prevent rubber gaskets, rubber cushions, and cock plug from coming out.
	LT-1B	790-129-9050	20 g (2 pcs.)	Polyethylene container	• Used in places requiring an immediately effective, strong adhesive. Used for plastics (except polyethylene, polyprophylene, tetrafluoroethlene and vinyl chloride), rubber, metal and non-metal.
	LT-2	09940-00030	50 g	Polyethylene container	• Features: Resistance to heat and chemicals • Used for anti-loosening and sealant purpose for bolts and plugs.
	LT-3	790-129-9060 (Set of adhesive and hardening agent)	Adhesive: 1 kg Hardening agent: 500 g	Can	• Used as adhesive or sealant for metal, glass and plastic.
	LT-4	790-129-9040	250 g	Polyethylene container	• Used as sealant for machined holes.
	Holtz MH 705	790-126-9120	75 g	Tube	• Used as heat-resisting sealant for repairing engine.
	Three bond 1735	790-129-9140	50 g	Polyethylene container	• Quick hardening type adhesive • Cure time: within 5 sec. to 3 min. • Used mainly for adhesion of metals, rubbers, plastics and woods.
	Aron-alpha 201	790-129-9130	2 g	Polyethylene container	• Quick hardening type adhesive • Quick cure type (max. strength after 30 minutes) • Used mainly for adhesion of rubbers, plastics and metals.
	Loctite 648-50	79A-129-9110	50 cc	Polyethylene container	• Resistance to heat, chemicals • Used at joint portions subject to high temperatures.
Gasket sealant	LG-1	790-129-9010	200 g	Tube	• Used as adhesive or sealant for gaskets and packing of power train case, etc.
	LG-5	790-129-9080	1 kg	Can	• Used as sealant for various threads, pipe joints, flanges. • Used as sealant for tapered plugs, elbows, nipples of hydraulic piping.
	LG-6	790-129-9020	200 g	Tube	• Features: Silicon based, resistance to heat, cold • Used as sealant for flange surface, tread. • Used as sealant for oil pan, final drive case, etc.
	LG-7	790-129-9070	1 kg	Tube	• Features: Silicon based, quick hardening type • Used as sealant for flywheel housing, intake manifold, oil pan, thermostat housing, etc.
	Three bond 1211	790-129-9090	100 g	Tube	• Used as heat-resisting sealant for repairing engine.
	Three bond 1207B	419-15-18131	100 g	Tube	• Features: Silicone type, heat resistant, vibration resistant, and impact resistant sealing material • Used as sealing material for transfer case

Category	Komatsu code	Part No.	Q'ty	Container	Main applications, features	
Molybdenum disulphide lubricant	LM-G	09940-00051	60 g	Can	• Used as lubricant for sliding portion (to prevent from squeaking).	
	LM-P	09940-00040	200 g	Tube	• Used to prevent seizure or scuffling of the thread when press fitting or shrink fitting. • Used as lubricant for linkage, bearings, etc.	
Grease	G2-LI	SYG2-400LI SYG2-350LI SYG2-400LI-A SYG2-160LI SYGA-160CNLI	Various	Various	• General purpose type	
	G2-CA	SYG2-400CA SYG2-350CA SYG2-400CA-A SYG2-160CA SYGA-160CNCA	Various	Various	• Used for normal temperature, light load bearing at places in contact with water or steam.	
	Molybdenum disulphide grease LM-G (G2-M)	SYG2-400M SYG2-400M-A SYGA-16CNM	400 g × 10 400 g × 20 16 kg	Bellows type Bellows type Can	• Used for heavy load portion	
	Hyper White Grease G2-T G0-T (*) *: For use in cold district	SYG2-400T-A SYG2-16CNT SYG0-400T-A (*) SYG0-16CNT (*)	400 g 16 kg	Bellows type Can	• Seizure resistance and heat resistance higher than molybdenum disulfide grease • Since this grease is white, it does not stand out against machine body.	
	Biogrease G2B G2-BT (*) *: For high temperature and large load	SYG2-400B SYGA-16CNB SYG2-400BT (*) SYGA-16CNBT (*)	400 g 16 kg	Bellows type Can	• Since this grease is decomposed by bacteria in short period, it has less effects on microorganisms, animals, and plants.	
Primer	SUNSTAR PAINT PRIMER 580 SUPER	417-926-3910	20 ml	Glass container	Adhesive for cab glass	• Used as primer for cab side (Using limit: 4 months)
	SUNSTAR GLASS PRIMER 580 SUPER		20 ml	Glass container		• Used as primer for glass side (Using limit: 4 months)
Adhesive	SUNSTAR PENGUINE SEAL 580 SUPER "S" or "W"		320 ml	Polyethylene container		• "S" is used for high-temperature season (April - October) and "W" for low-temperature season (November - April) as adhesive for glass. (Using limit: 4 months)
	Sika Japan, Sikaflex 256HV	20Y-54-39850	310 ml	Polyethylene container		• Used as adhesive for glass. (Using limit: 6 months)
Caulking material	SUNSTAR PENGUINE SEAL No. 2505	417-926-3920	320 ml	Polyethylene container		• Used to seal joints of glass parts. (Using limit: 4 months)
	SEKISUI SILICONE SEALANT	20Y-54-55130	333 ml	Polyethylene container		• Used to seal front window. (Using limit: 6 months)

STANDARD TIGHTENING TORQUE

STANDARD TIGHTENING TORQUE TABLE (WHEN USING TORQUE WRENCH)

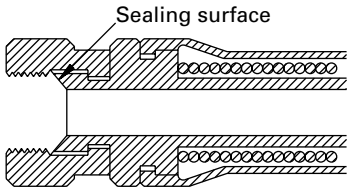
★ In the case of metric nuts and bolts for which there is no special instruction, tighten to the torque given in the table below.

Thread diameter of bolt	Width across flats	Tightening torque	
		 CDL00372	
mm	mm	Nm	kgm
6	10	11.8 – 14.7	1.2 – 1.5
8	13	27 – 34	2.8 – 3.5
10	17	59 – 74	6 – 7.5
12	19	98 – 123	10 – 12.5
14	22	153 – 190	15.5 – 19.5
16	24	235 – 285	23.5 – 29.5
18	27	320 – 400	33 – 41
20	30	455 – 565	46.5 – 58
22	32	610 – 765	62.5 – 78
24	36	785 – 980	80 – 100
27	41	1150 – 1440	118 – 147
30	46	1520 – 1910	155 – 195
33	50	1960 – 2450	200 – 250
36	55	2450 – 3040	250 – 310
39	60	2890 – 3630	295 – 370

Thread diameter of bolt	Width across flats	Tightening torque	
		 CDL00373	
mm	mm	Nm	kgm
6	10	5.9 – 9.8	0.6 – 1.0
8	13	13.7 – 23.5	1.4 – 2.4
10	14	34.3 – 46.1	3.5 – 4.7
12	27	74.5 – 90.2	7.6 – 9.2

TABLE OF TIGHTENING TORQUES FOR FLARED NUTS

★ In the case of flared nuts for which there is no special instruction, tighten to the torque given in the table below.



SAD00483

Thread diameter	Width across flat	Tightening torque	
mm	mm	Nm	kgm
14	19	24.5 ± 4.9	2.5 ± 0.5
18	24	49 ± 19.6	5 ± 2
22	27	78.5 ± 19.6	8 ± 2
24	32	137.3 ± 29.4	14 ± 3
30	36	176.5 ± 29.4	18 ± 3
33	41	196.1 ± 49	20 ± 5
36	46	245.2 ± 49	25 ± 5
42	55	294.2 ± 49	30 ± 5

TABLE OF TIGHTENING TORQUES FOR SPLIT FLANGE BOLTS

★ In the case of split flange bolts for which there is no special instruction, tighten to the torque given in the table below.

Thread diameter	Width across flat	Tightening torque	
mm	mm	Nm	kgm
10	14	59 – 74	6 – 7.5
12	17	98 – 123	10 – 12.5
16	22	235 – 285	23.5 – 29.5

TABLE OF TIGHTENING TORQUES FOR O-RING BOSS PIPING JOINTS

★ Unless there are special instructions, tighten the O-ring boss piping joints to the torque below.

Norminal No.	Thread diameter	Width across flat	Tightening torque (Nm {kgm})	
	mm	mm	Range	Target
02	14	Varies depending on type of connector.	35 – 63 {3.5 – 6.5}	44 {4.5}
03, 04	20		84 – 132 {8.5 – 13.5}	103 {10.5}
05, 06	24		128 – 186 {13.0 – 19.0}	157 {16.0}
10, 12	33		363 – 480 {37.0 – 49.0}	422 {43.0}
14	42		746 – 1010 {76.0 – 103}	883 {90.0}

TABLE OF TIGHTENING TORQUES FOR O-RING BOSS PLUGS

★ Unless there are special instructions, tighten the O-ring boss plugs to the torque below.

Norminal No.	Thread diameter	Width across flat	Tightening torque (Nm {kgm})	
	mm	mm	Range	Target
08	08	14	5.88 – 8.82 {0.6 – 0.9}	7.35 {0.75}
10	10	17	9.8 – 12.74 {1.0 – 1.3}	11.27 {1.15}
12	12	19	14.7 – 19.6 {1.5 – 2.0}	17.64 {1.8}
14	14	22	19.6 – 24.5 {2.0 – 2.5}	22.54 {2.3}
16	16	24	24.5 – 34.3 {2.5 – 3.5}	29.4 {3.0}
18	18	27	34.3 – 44.1 {3.5 – 4.5}	39.2 {4.0}
20	20	30	44.1 – 53.9 {4.5 – 5.5}	49.0 {5.0}
24	24	32	58.8 – 78.4 {6.0 – 8.0}	68.6 {7.0}
30	30	32	93.1 – 122.5 {9.5 – 12.5}	107.8 {11.0}
33	33	—	107.8 – 147.0 {11.0 – 15.0}	124.4 {13.0}
36	36	36	127.4 – 176.4 {13.0 – 18.0}	151.9 {15.5}
42	42	—	181.3 – 240.1 {18.5 – 24.5}	210.7 {21.5}
52	52	—	274.4 – 367.5 {28.0 – 37.5}	323.4 {33.0}

TIGHTENING TORQUE FOR 102 ENGINE SERIES

1) BOLT AND NUTS

Use these torques for bolts and nuts (unit: mm) of Cummins Engine.

Thread diameter	Tightening torque	
mm	Nm	kgm
6	10 ± 2	1.02 ± 0.20
8	24 ± 4	2.45 ± 0.41
10	43 ± 6	4.38 ± 0.61
12	77 ± 12	7.85 ± 1.22

2) EYE JOINTS

Use these torques for eye joints (unit: mm) of Cummins Engine.

Thread diameter	Tightening torque	
mm	Nm	kgm
6	8 ± 2	0.81 ± 0.20
8	10 ± 2	1.02 ± 0.20
10	12 ± 2	1.22 ± 0.20
12	24 ± 4	2.45 ± 0.41
14	36 ± 5	3.67 ± 0.51

3) TAPERED SCREWS

Use these torques for tapered screws (unit: inch) of Cummins Engine.

Thread diameter	Tightening torque	
inch	Nm	kgm
1 / 16	3 ± 1	0.31 ± 0.10
1 / 8	8 ± 2	0.81 ± 0.20
1 / 4	12 ± 2	1.22 ± 0.20
3 / 8	15 ± 2	1.53 ± 0.41
1 / 2	24 ± 4	2.45 ± 0.41
3 / 4	36 ± 5	3.67 ± 0.51
1	60 ± 9	6.12 ± 0.92

TIGHTENING TORQUE TABLE FOR HOSES (TAPER SEAL TYPE AND FACE SEAL TYPE)

★ Tighten the hoses (taper seal type and face seal type) to the following torque, unless otherwise specified.

★ Apply the following torque when the threads are coated (wet) with engine oil.

Nominal size of hose	Width across flats	Tightening torque (Nm {kgm})		Taper seal type	Face seal type	
		Range	Target	Thread size (mm)	Nominal thread size - Threads per inch, Thread series	Root diameter (mm) (Reference)
02	19	34 – 54 {3.5 – 5.5}	44 {4.5}	–	$\frac{9}{16}$ – 18UN	14.3
		34 – 63 {3.5 – 6.5}	44 {4.5}	14	–	–
03	22	54 – 93 {5.5 – 9.5}	74 {7.5}	–	$\frac{11}{16}$ – 16UN	17.5
	24	59 – 98 {6.0 – 10.0}	78 {8.0}	18	–	–
04	27	84 – 132 {8.5 – 13.5}	103 {10.5}	22	$\frac{13}{16}$ – 16UN	20.6
05	32	128 – 186 {13.0 – 19.0}	157 {16.0}	24	1 – 14UNS	25.4
06	36	177 – 245 {18.0 – 25.0}	216 {22.0}	30	$1\frac{3}{16}$ – 12UN	30.2
(10)	41	177 – 245 {18.0 – 25.0}	216 {22.0}	33	–	–
(12)	46	197 – 294 {20.0 – 30.0}	245 {25.0}	36	–	–
(14)	55	246 – 343 {25.0 – 35.0}	294 {30.0}	42	–	–

ELECTRIC WIRE CODE

In the wiring diagrams, various colors and symbols are employed to indicate the thickness of wires.
This wire code table will help you understand WIRING DIAGRAMS.
Example: 5WB indicates a cable having a nominal number 5 and white coating with black stripe.

CLASSIFICATION BY THICKNESS

Norminal number	Copper wire			Cable O.D. (mm)	Current rating (A)	Applicable circuit
	Number of strands	Dia. of strands (mm^2)	Cross section (mm^2)			
0.85	11	0.32	0.88	2.4	12	Starting, lighting, signal etc.
2	26	0.32	2.09	3.1	20	Lighting, signal etc.
5	65	0.32	5.23	4.6	37	Charging and signal
15	84	0.45	13.36	7.0	59	Starting (Glow plug)
40	85	0.80	42.73	11.4	135	Starting
60	127	0.80	63.84	13.6	178	Starting
100	217	0.80	109.1	17.6	230	Starting

CLASSIFICATION BY COLOR AND CODE

Priority	Classification \ Circuits		Charging	Ground	Starting	Lighting	Instrument	Signal	Other
1	Primary	Code	W	B	B	R	Y	G	L
		Color	White	Black	Black	Red	Yellow	Green	Blue
2	Auxiliary	Code	WR	—	BW	RW	YR	GW	LW
		Color	White & Red	—	White & Black	Red & White	Rellow & Red	Green & White	Blue & White
3		Code	WB	—	BY	RB	YB	GR	LR
		Color	White & Black	—	Black & Yellow	Red & Black	Yellow & Black	Green & Red	Blue & Yellow
4		Code	WL	—	BR	RY	YG	GY	LY
		Color	White & Blue	—	Black & Red	Red & Yellow	Yellow & Green	Green & Yellow	Blue & Yellow
5		Code	WG	—	—	RG	YL	GB	LB
		Color	White & Green	—	—	Red & Green	Yellow & Blue	Green & Black	Blue & Black
6		Code	—	—	—	RL	YW	GL	—
		Color	—	—	—	Red & Blue	Yellow & White	Green & Blue	—

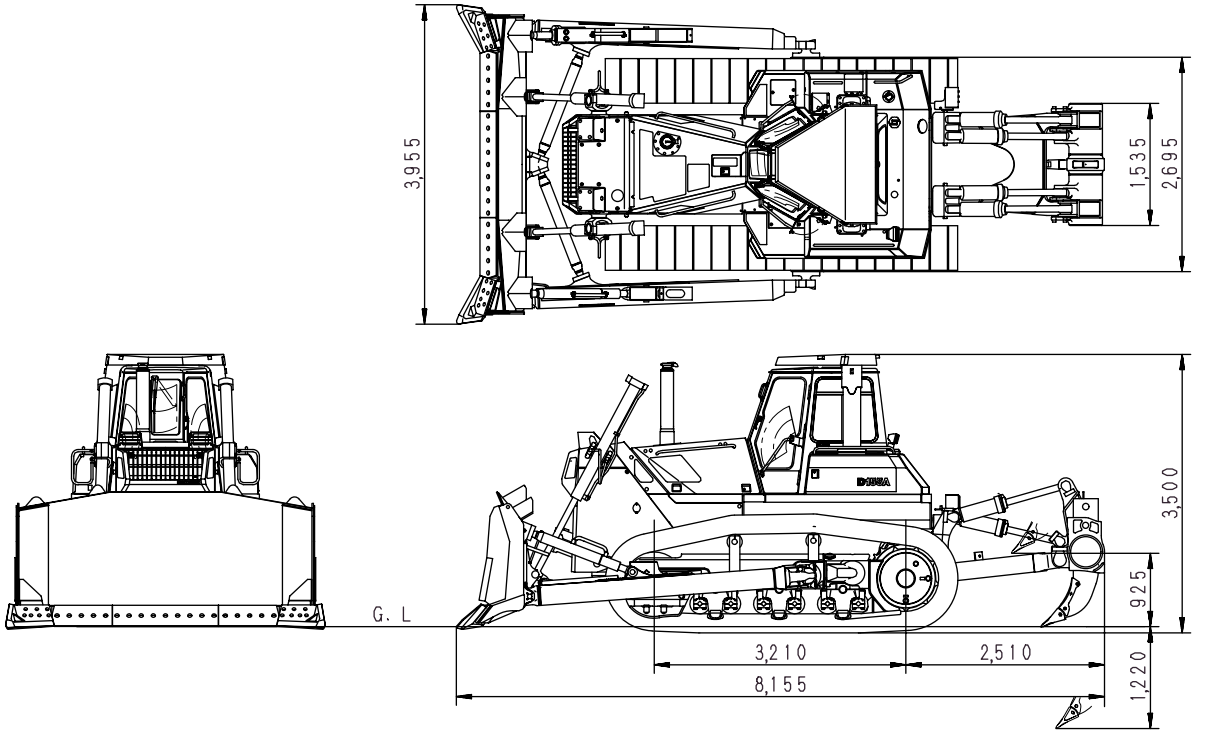
01 General

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Specification drawing

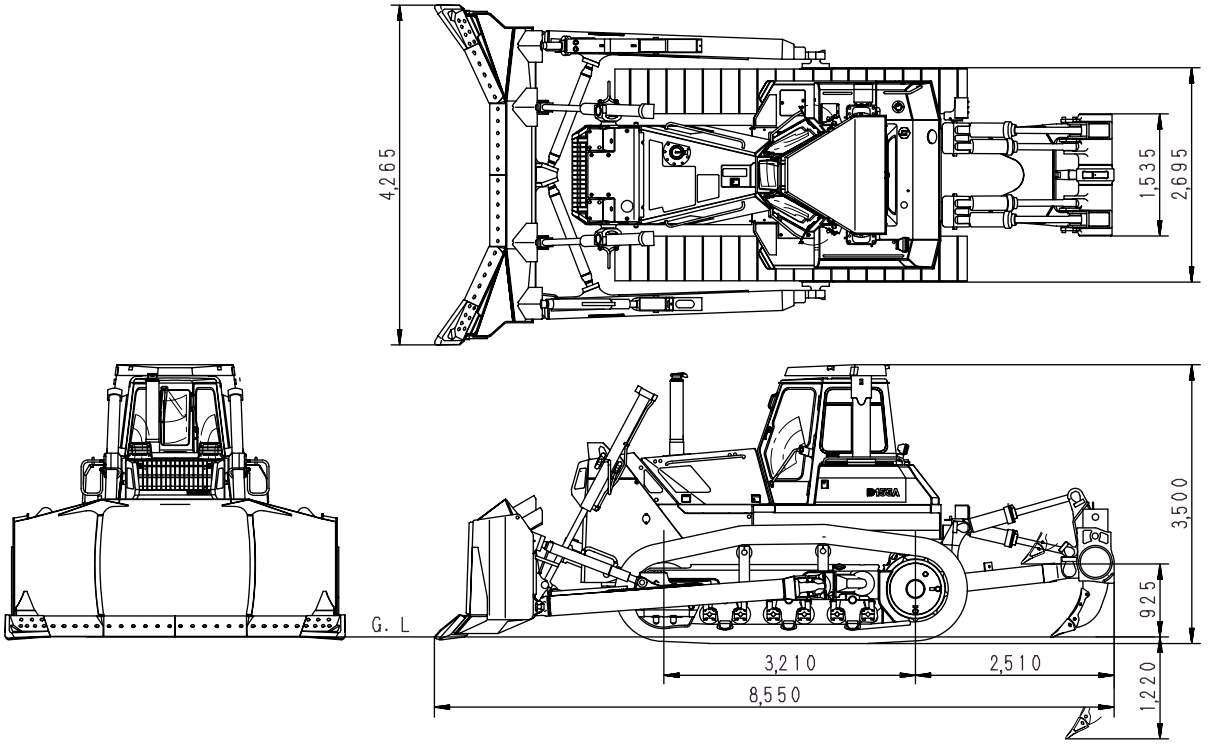
- Semi U-tiltdozer + giant ripper

Unit: mm



SXD04070

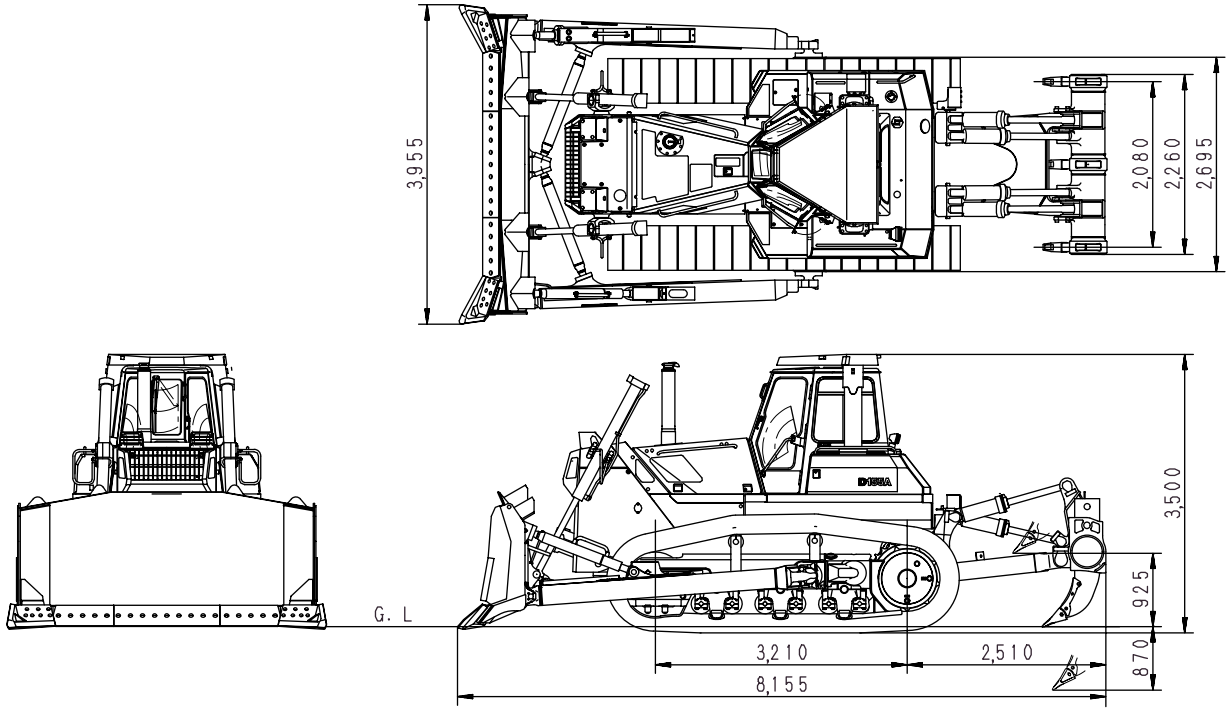
- U-tiltdozer + giant ripper



SXD04071

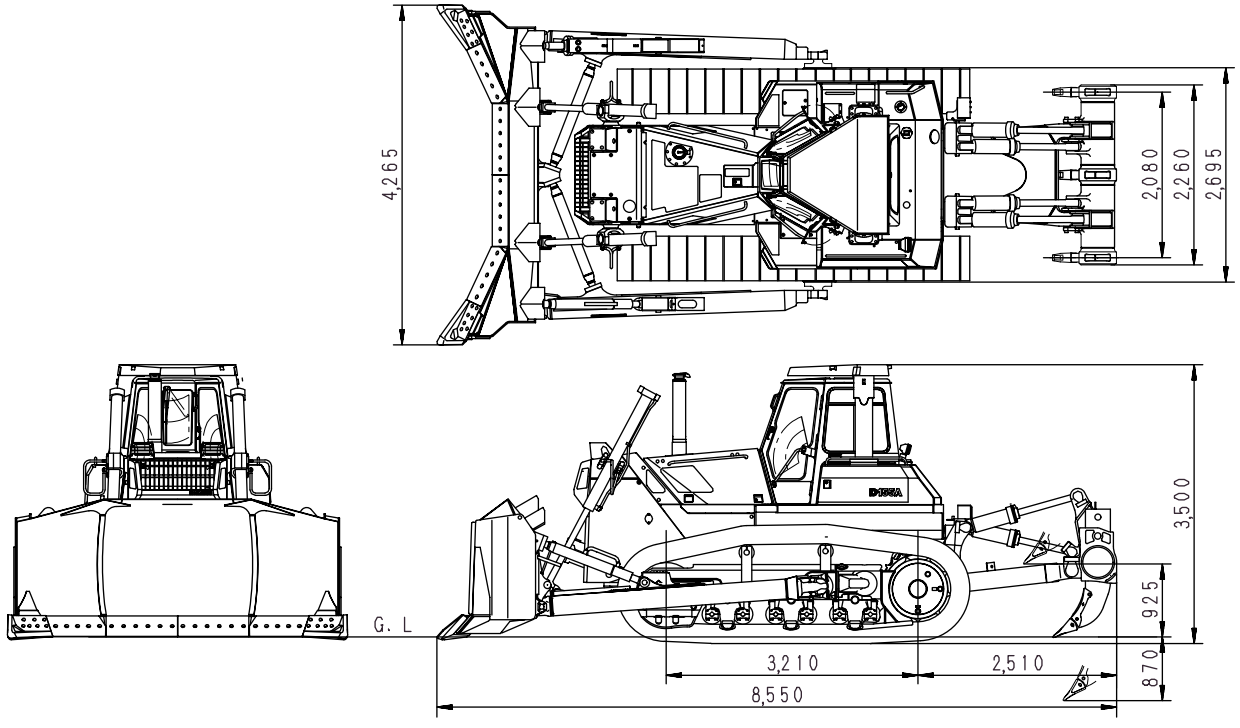
- Semi U-tiltdozer + multi-shank ripper

Unit: mm



SXD04072

- U-tiltdozer + multi-shank ripper



SXD04073

Specifications

Machine model				D155A-5
Serial Number				65001 and up
Operating weight	Bare tractor		kg	27,900
	With semi U-dozer + variable multi-shank ripper + ROPS + cab + air conditioner + side cover			38,000
Performance	Minimum turning radius (center of machine)		m	1.05
	Gradeability		deg	30
	Stability (front, rear, left, right)		deg	35
	Transmission speed ranges: Forward	1st	km/h	3.7
		2nd		6.7
		3rd		11.0
	Transmission speed ranges: Reverse	1st	km/h	5.0
		2nd		8.2
		3rd		13.9
	Ground pressure	Bare tractor	kPa {kg/cm^2}	76.48 {0.78}
		With semi U-dozer + variable multi-shank ripper + ROPS + cab + air conditioner + side cover		103.94 {1.06}
		With U-dozer + variable multi-shank ripper + ROPS + cab + air conditioner + side cover		106.88 {1.09}
Dimensions	Overall length	Bare tractor	mm	4,975
		With semi U-dozer + variable multi-shank ripper		8,155
		With U-dozer + variable multi-shank ripper		8,550
	Overall width	Bare tractor	mm	2,695
		With semi U-dozer + variable multi-shank ripper		3,955
		With U-dozer + variable multi-shank ripper		4,265
	Overall height	To top of exhaust pipe	mm	3,395
		To top of cab (to top of control lever)		2,590

Machine model				D155A-5
Serial Number				65001 and up
Dimensions	Overall height	With cab installed	mm	3,365
		With ROPS canopy installed		3,500
		With canopy installed		3,500
	Track gauge		mm	2,100
	Length of track on ground (mm)			3,210
	Track shoe width (standard)			560
	Minimum ground clearance (To bottom surface of undercover)			485
Engine	Name			SA6D140E-2
	Type of engine			4-cycle, water-cooled, in-line, vertical type, direct injection, with turbocharger and aftercooler
	No. of cylinders - Bore x Stroke		mm	6-140 x 165
	Piston displacement		ℓ {cc}	15.24 {15,240}
	Performance	Flywheel horsepower	kW{HP}/rpm	225{301}/1,900
		Max. torque	Nm{kgm}/rpm	1,432{146}/1,250
		Max. speed at no load	rpm	2,050
		Min. speed at no load	rpm	700
		Min. fuel consumption ratio	g/kWh{g/HPh}	205 {153}
	Starting motor			24V, 11kW
	Alternator			24V, 35A
	Battery			12V, 170Ah x 2
	Radiator core type			D-7
Power train	Torque converter			3-element, single-stage, single phase (TCS43-17A)
	Transmission			Planetary gear, multiple disc clutch, hydraulically actuated, force-feed lubrication gear pump, forward 3-speed, reverse 3-speed, manual mechanical type

Machine model			D155A-5
Serial Number			65001 and up
Power train	Bevel gear shaft		Spiral bevel gear, force-feed lubrication gear pump
	Clutch and brake	Steering clutch	Wet type, multiple clutch disc, spring boosted, manual type, hydraulically actuated
		Steering brake	Wet type, multiple clutch disc, spring boosted, interconnected with clutch, pedal operated, manually operated, hydraulically actuated
	Final drive		Spur gear 1-stage, planetary gear 1-stage, splash type lubrication
Undercarriage	Suspension		Rigid, balancing beam type
	Carrier roller		2 each side
	Track roller		6 each side
	Track shoe (standard shoe width: 560 mm)		Assembly type, single grouser, 41 each side, pitch: 228.6 mm
Power train + lubrication pump (tandem)			Gear type (SAL(3)80+100)
Scavenging pump			Gear type (112)
PPC charge pump (tandem with hydraulic pump (SAR(4)112))			Gear type (SAR(1)10) Max. delivery pressure : 3.7 MPa {38 kg/cm^2} Theoretical delivery : 22 ℓ/min/2,223 rpm
Work equipment hydraulic system	Hydraulic pump (tandem)		Gear type (SAR(4)112) Max. delivery pressure : 20.6 MPa {210 kg/cm^2} Theoretical delivery : 249 ℓ/min/2,223 rpm
	Main control valve	Type • For blade lift • For blade tilt • For ripper lift • For ripper tilt	2+1+1 tandem spool type, hydraulically assisted type Ripper tilt + ripper lift + blade lift + blade tilt

Machine model					D155A-5
Serial Number					65001 and up
Type					Reciprocal, piston type
Work equipment hydraulic system	Hydraulic cylinder	Dimensions of blade lift cylinder	Cylinder bore	mm	120
			Piston rod outer diameter		75
			Max. piston stroke		1,347
			Max. distance between pins		3,187
			Min. distance between pins		1,836
		Dimensions of blade tilt cylinder	Cylinder bore	mm	180
			Piston rod outer diameter		90
			Max. piston stroke		160
			Max. distance between pins		1,468
			Min. distance between pins		1,308
		Dimensions of blade tilt, pitch cylinder	Cylinder bore	mm	200
			Piston rod outer diameter		90
			Max. piston stroke		160
			Max. distance between pins		1,468
			Min. distance between pins		1,308
		Dimensions of ripper lift cylinder	Cylinder bore	mm	160
			Piston rod outer diameter		80
			Max. piston stroke		510
			Max. distance between pins		1,538
			Min. distance between pins		1,028
		Dimensions of ripper tilt cylinder	Cylinder bore	mm	160
			Piston rod outer diameter		80
			Max. piston stroke		350
			Max. distance between pins		1,545
			Min. distance between pins		1,195

Machine model					D155A-5
Serial Number					65001 and up
Work equipment	Hydraulic tank				Box type (control valve externally installed)
	Type				Hydraulic semi U-tiltdozer, hydraulic U-tiltdozer
	Blade support method				Brace type (right tilt cylinder)
	Semi U-dozer	Performance	Blade lifting height (from ground level)	mm	1,250
			Blade lowering depth (from ground level)	mm	590
			Max. blade tilt	mm	1,000
			Blade cutting angle variation	deg	± 6
			Blade capacity	m^3	8.8 < SAE 11.7 >
		Dimensions	Blade width	mm	3,955
			Blade height	mm	1,720
			Blade cutting angle	deg	52
	U-dozer	Performance	Blade lifting height (from ground level)	mm	1,250
			Blade lowering depth (from ground level)	mm	590
			Max. blade tilt	mm	1,080
			Blade cutting angle variation	deg	± 6
			Blade capacity	m^3	11.8 < SAE 13.2 >
		Dimensions	Blade width	mm	4,265
			Blade height	mm	1,720
			Blade cutting angle	deg	52
	Semi U-dozer + pitch	Performance	Max. tilt	mm	1,000
			Max. pitch angle	deg	6
			Max. pitch-back angle	deg	6
	U-dozer + pitch	Performance	Max. tilt	mm	1,080
			Max. pitch angle	deg	6
			Max. pitch-back angle	deg	6

Machine model					D155A-5
Serial Number					65001 and up
Work equipment	Multi-shank ripper	Performance	Max. digging depth	mm	870
			Max. lifting height	mm	925
			Ripping angle of point	deg	34.5 – 60
		Dimensions	Max. beam width	mm	2,260
			Shank positions	No. of holes	2
	Giant ripper	Performance	Max. digging depth	mm	1,220
			Max. lifting height	mm	925
			Ripping angle of point	deg	34.5 – 60
		Dimensions	Max. beam width	mm	1,535
			Shank positions	No. of holes	3

Weight table

⚠ This weight table is a guide for use when transporting or handling components. Unit: kg

Machine model	D155A-5
Serial Number	65001 and up
Engine, damper assembly	1,704
• Engine assembly	1,556
• Damper assembly	117
• Universal joint	18
• Engine mount parts (wiring)	3
Radiator assembly (including built-in oil cooler)	305
Hydraulic cooler	34 (15 as individual part)
Fuel tank assembly (when empty)	235
Fuel tank assembly (when full)	660
Power train unit assembly	2,188
• Torque converter, PTO assembly	319
• Transmission assembly	645
• Steering clutch, brake assembly	995
• Steering valve assembly	25
• Transmission control valve assembly	17
• Control valve cover assembly	13
• Main relief valve assembly	5
• PPC charge valve assembly	7
• Power train filter assembly	20
• Scavenging pump	17
• Power train, lubrication pump	27
• Hydraulic, PPC charge pump	25
Final drive assembly	1,372 x 2
Sprocket teeth	(12 x 9) x 2
Hull frame assembly	3,182
Track group assembly (each side)	3,755
• Track frame	1,224
• Idler assembly	410
• Recoil spring assembly	508
• Bogie, track roller assembly	410 x 3
• Track roller assembly (single flange x1)	97 x 3
• Track roller assembly (double flange x1)	109 x 3
• Carrier roller assembly	45 x 2

Unit: kg

Machine model	D155A-5
Serial Number	65001 and up
Track shoe assembly (560 mm, wet type)	2,558 x 2
Pivot shaft assembly	116 x 2
Equalizer bar	229
Hydraulic tank assembly	102
Hydraulic filter assembly	19
Main control valve	99
(Blade lift + blade tilt + ripper lift + ripper tilt)	
Engine underguard	183
Transmission underguard	228
Operator's seat	42
Semi U-tiltdozer assembly	4,903
• Blade	2,818
• Straight frame	690 x 2
• Tilt brace	98 x 1
• Center brace	92 x 2
• Tilt cylinder assembly	190
Semi U-tilt + pitch dozer assembly	5,007
• Blade	2,818
• Straight frame	690 x 2
• Center brace	92 x 2
• Tilt cylinder assembly	190
• Pitch cylinder assembly	215
Blade lift cylinder assembly	165 x 2
Multi-shank ripper assembly	3,600
• Bracket, beam	2,014
• Shank	266 x 3
• Lift cylinder assembly	119 x 2
• Tilt cylinder assembly	122 x 2
Giant ripper assembly	3,135
• Bracket, beam	1,728
• Shank	333 x 1
• Lift cylinder assembly	119 x 2
• Tilt cylinder assembly	122 x 2
• Pin puller cylinder assembly	7
ROPS assembly	480
Cab	285
ROPS canopy	698
Canopy	105
Dashboard	83
Floor frame	310

Table of fuel, coolant and lubricants

Reservoir	Kind of fluid	Ambient temperature	Capacity (ℓ)	
		–22 –4 14 32 50 68 86 104°F –30 –20 –10 0 10 20 30 40°C	Specified	Refille
Engine oil pan	Engine oil	SAE 30 / SAE 10W / SAE 10W-30 / SAE 15W-40	42	37
Power train case	Engine oil	SAE 30	105	60
Final drive case (each side)	Engine oil	SAE 10W	58	58
Damper case	Engine oil	SAE 30	1.5	1.5
Pivot shaft case (each side)	Engine oil	SAE 30	8.7	8.7
Hydraulic system	Engine oil	SAE 10W / SAE 10W-30 / SAE 15W-40	126 (w/o attachment) 190 (with blade, ripper)	97 97
Idler (each)	Gear oil	GO 140	0.59	0.59
Track roller (each)	Gear oil	GO 140	0.45	0.45
Carrier roller (each)	Gear oil	GO 140	0.68	0.68
Bogie (each)	Gear oil	GO 140	0.21 – 0.23	0.21 – 0.23
Fuel tank	Diesel fuel	ASTM D975 No. 2 / ASTM D975 No.1	500	—
Cooling system (including reservoir tank)	Coolant	Add antifreeze	97	—

Note:

(1) When fuel sulphur content is less than 0.5 %, change oil in the oil pan every periodic maintenance hours described in this manual.
Change oil according to the following table if fuel sulphur content is above 0.5 %.

Fuel sulphur content	Change interval of oil in engine oil pan
0.5 to 1.0 %	1/2 of regular interval
Above 1.0 %	1/4 of regular interval

ASTM: American Society of Testing and Material
SAE: Society of Automotive Engineers
API: American Petroleum Institute

(2) When starting the engine in an atmospheric temperature of lower than 0°C, be sure to use engine oil of SAE10W, SAE10W-30 and SAE15W-40 even though an atmospheric temperature goes up to 10°C more or less in the day time.

(3) Use API classification CD as engine oil and if API classification CC, reduce the engine oil change interval to half.

(4) There is no problem if single grade oil is mixed with multigrade oil (SAE10W-30, 15W-40), but be sure to add single grade oil that matches the temperature in the table on the left.

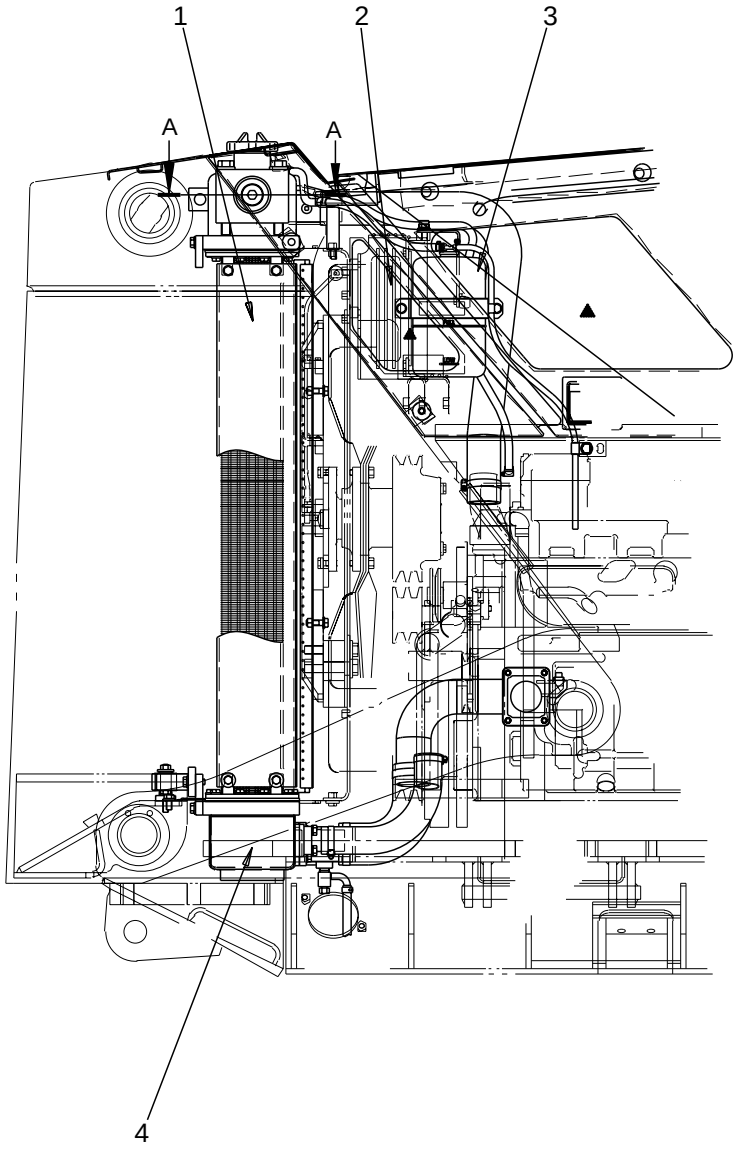
(5) We recommend Komatsu genuine oil which has been specifically formulated and approved for use in engine and hydraulic work equipment applications.

Specified capacity: Total amount of oil including oil for components and oil in piping.
Refill capacity: Amount of oil needed to refill system during normal inspection and maintenance.

10 Structure and function

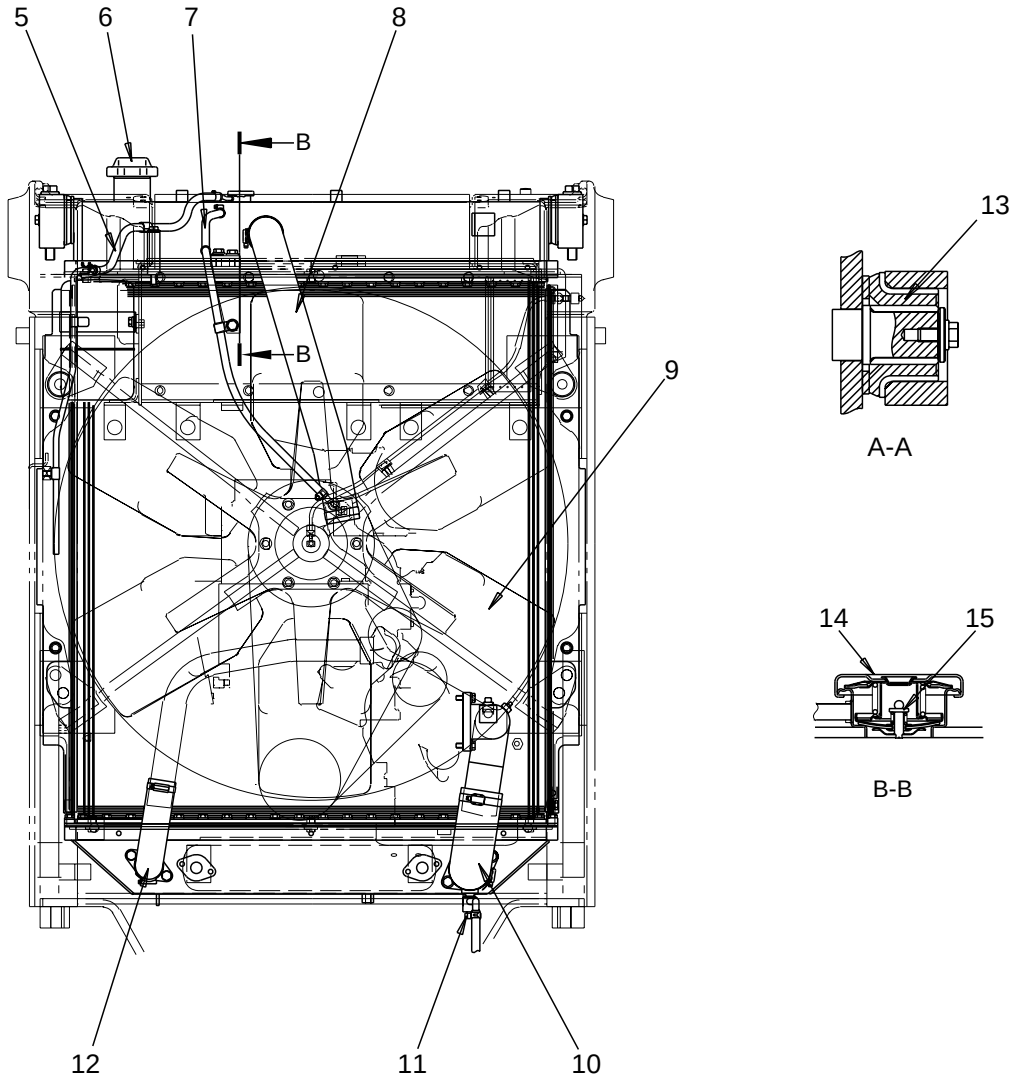
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Radiator



SKD00002

1. Radiator core assembly
2. Fan guard
3. Reservoir tank
4. Lower tank (power train oil cooler built-in)
5. Overflow hose
6. Water filler cap
7. Air bleed hose
8. Inlet hose
9. Fan
10. Outlet hose
11. Drain plug
12. Inlet hose (for circulating at low water temperature)
13. Cushion
14. Breather cap
15. Valve



SKD00003

Outline

- The radiator is provided with a reservoir tank to make it easier to check and maintain the radiator water level.
- The power train oil cooler is built into the lower tank.

			Radiator
Core type			D-7
Fin pitch		(mm)	4.0
Heat dissipation area		(m^2)	102.42
Pressure valve	Relief pressure	(MPa {kg/cm^2})	0.09 (0.9)
	Vacuum pressure	(MPa {kg/cm^2})	0.005 (0.05)



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